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Please digest these

I've digested both documents. Here is a concise technical synthesis, emphasizing the concepts most relevant to your interests in particle impacts, cold spray, dispersion strengthening, and high-temperature structural materials.

Paper 1: Creep Behavior of Particle-Strengthened Alloys (ODS and precipitate-strengthened alloys)

Source: Chapter 8, *Fundamentals of Creep in Metals and Alloys*  kassner-ODS

Central Question

Why do oxide dispersion strengthened (ODS) alloys and particle-strengthened alloys retain strength during creep far beyond what classical Orowan strengthening predicts?

The chapter focuses on:

- ODS alloys (incoherent particles)
- Low-volume-fraction precipitate-strengthened alloys (<10–30%)
- Threshold stress behavior during creep

Key Concept 1: Orowan Strengthening is Not Enough

At low temperatures, strengthening is governed by:

$$\tau_{or} \approx \frac{Gb}{L}$$

where:

- G = shear modulus
- b = Burgers vector

- L = particle spacing

Particles force dislocations to bow between obstacles and leave Orowan loops.

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However, at elevated temperatures dislocations can climb around particles, meaning Orowan bowing alone cannot explain the creep resistance.

Key Concept 2: Threshold Stress

ODS alloys exhibit a stress below which creep becomes extremely slow:

$$\sigma_{eff} = \sigma - \sigma_{th}$$

This threshold stress is observed experimentally in Ni–Cr–ThO₂ and many ODS systems.

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The creep law becomes:

$$\dot{\epsilon} = A(\sigma - \sigma_{th})^n$$

instead of

$$\dot{\epsilon} = A\sigma^n$$

This is one of the defining features of dispersion-strengthened alloys.

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Key Concept 3: Two Main Explanations for Threshold Stress

A. Dislocation Climb Resistance

Dislocations climb around particles.

Climbing increases dislocation line length:

$$dE \approx \frac{Gb^2}{2} dL - \tau b L dx$$

The added line energy creates an apparent threshold stress.

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Two limiting cases:

Local Climb

Climb restricted near particle interfaces.

Predicted threshold:

$$\sigma_c \approx 0.19 \frac{Gb}{L}$$

General Climb

Climb extends into matrix.

Predicted threshold:

$$\sigma_c \approx 0.004 - 0.02 \frac{Gb}{L}$$

Much smaller.  kassner-ODS

B. Dislocation Detachment Model

TEM observations show dislocations become trapped at particle interfaces.

After climbing around a particle, the dislocation must detach from the interface.

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Detachment stress:

$$\tau_d = \sqrt{1 - k_R^2} \left(\frac{Gb}{L} \right)$$

where k_R describes elastic energy relaxation at the particle interface.  kassner-ODS

This predicts:

$$\tau_d \approx \frac{Gb}{3L}$$

which is substantially larger than general-climb predictions.

Today, detachment-controlled creep is considered one of the most important explanations for ODS creep resistance.

Key Concept 4: Thermally Activated Detachment

Rosler and Arzt developed a kinetic model where detachment is thermally activated.

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Important consequence:

The threshold stress is not truly absolute.

Instead:

- creep can still occur below the apparent threshold
- rate becomes exponentially small

This better matches experiments.

Key Concept 5: Optimum Particle Size

The detachment model predicts an optimum particle size:

Too small:

- detachment easier

Too large:

- Orowan stress decreases

Thus maximum creep resistance occurs at an intermediate particle size.

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This is extremely relevant for:

- ODS superalloys
 - ODS steels
 - advanced space nuclear materials
-

Key Concept 6: Coherent vs Incoherent Particles

Incoherent particles

Examples:

- Y_2O_3 dispersoids
- ThO_2 dispersoids

Mechanism:

- climb
- detachment
- interfacial pinning

Coherent particles

Examples:

- γ' in superalloys
- Al_3Sc

Can initially strengthen by:

- particle cutting
- APB formation
- coherency stresses

At elevated temperatures many eventually become climb-controlled.

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Most Important Takeaway

For high-temperature applications:

Particle spacing and interface characteristics matter more than simple Orowan strengthening.

The dominant creep resistance mechanism often becomes:

climb + interfacial detachment

rather than pure bowing.

This is directly relevant to:

- ODS alloys
 - cold spray deposits containing oxide dispersoids
 - refractory alloy design
 - hypersonic materials
-

Paper 2: Effects of Precipitate Shape and Orientation on Dispersion Strengthening in Mg Alloys

Nie (2003)  1-s2.0-S1359646202004979-main

Central Question

For the same:

- precipitate volume fraction
- precipitate number density

which precipitate shape/orientation gives the greatest strengthening?

Key Result

Not all precipitates are equally effective.

The strengthening effectiveness depends strongly on:

1. Shape
2. Aspect ratio
3. Crystallographic orientation

not simply volume fraction.  1-s2.0-S1359646202004979-main

Orowan Framework

Strengthening:

$$\Delta\tau = \frac{Gb}{2\pi\lambda} \ln \left(\frac{d_p}{r_0} \right)$$

where λ is obstacle spacing.  1-s2.0-S1359646202004979-main

Reducing λ is the key to strengthening.

Comparison of Precipitate Types

The paper evaluates:

1. Spherical particles

Reference case.

2. Basal plates

Plates lying parallel to the Mg basal plane.

Result:

Weak strengthening.

Increasing aspect ratio actually reduces effectiveness because the number of obstacles intersecting the slip plane decreases.  1-s2.0-S1359646202004979-main

3. [0001] Rods

Moderately effective.

Better than spheres.

Better than basal plates.  1-s2.0-S1359646202004979-main

4. Prismatic Plates

Most effective by far.

These plates intersect the basal slip plane nearly perpendicularly.

Dislocations encounter many more barriers.  1-s2.0-S1359646202004979-main

Strengthening Ranking

For identical volume fraction and particle density:

Prismatic plates > [0001] rods > spheres > basal plates

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Aspect Ratio Effects

Prismatic plates

Strength increases dramatically with aspect ratio.

For large aspect ratios:

$$\lambda \rightarrow 0$$

meaning the effective spacing between obstacles approaches zero.

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The model predicts theoretically infinite Orowan strengthening if plates become continuous barriers.

Of course, in reality:

- local stresses rise
- precipitates shear
- other mechanisms activate

before infinity is reached.

Experimental Connection

The paper explains why:

AZ91 (Mg-Al-Zn)

Strengthened mainly by basal $\text{Mg}_{17}\text{Al}_{12}$ plates

=

 ...

WE54 (Mg-Al-Zn)

Strengthened by prismatic β_1/β plates

→ much higher strength (~200 MPa).

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Combined Insights Relevant to Your Research

Several ideas connect strongly to your work on cold spray, impact bonding, and extreme-environment alloys:

1. Interface-Controlled Strengthening

The ODS chapter shows that particle interfaces themselves can dominate high-temperature deformation through dislocation trapping and detachment.

This may be highly relevant for:

- oxide films in cold spray
 - oxide fragments at particle-particle interfaces
 - impact-generated nanostructures
-

2. Geometry Matters as Much as Volume Fraction

The Mg paper demonstrates that orientation and aspect ratio can dominate strengthening.

For cold spray:

- elongated oxides
- lamellar oxide remnants
- impact-generated intermetallic layers

may provide much different strengthening than equivalent spherical particles.

3. High-Temperature Design

For aerospace and hypersonics:

The ideal strengthening architecture is not necessarily:

- maximum particle fraction

but rather:

- optimized spacing
 - optimized aspect ratio
 - strong particle/matrix interfaces
 - resistance to dislocation detachment
-

4. Potential Relevance to Regolith and Space Manufacturing

The detachment/interfacial-pinning framework may provide a useful conceptual model for:

- oxide-rich extraterrestrial alloys,
- lunar regolith-derived metals,
- in-space cold spray deposits,

where nonmetallic inclusions and interfaces could act similarly to ODS dispersoids.

In short, the first document is fundamentally about **why particles improve creep resistance at high temperature**, while the second is about **how particle shape and crystallographic orientation determine Orowan strengthening efficiency**. Together they provide a useful framework linking particle geometry, interface physics, and elevated-temperature mechanical behavior.







 Sources

Okay, I have a student, Denise, who has selected data related to ODS alloys in terms of particle size, volume fraction for various alloys, interparticle spacing, etc. She is also on track to collect a lot of creep-related data for each of these alloys. We want to design a paper where we describe different kinds of Ashby plots for these ODS alloys based on some of the resources I've given you, the two papers, but also other papers. Give me at least 10 Ashby plots or more highlighting what is it that I should be plotting along X and Y axis. We want this paper to essentially be the ultimate design paper for ODS alloys

Show more ▾

Thought for a few seconds >

Below are **15 high-value Ashby-style plots** Denise can build for an “ultimate design map” paper on ODS alloys, including creep, particle geometry, interfacial pinning, and comparisons to faceted/plate-like precipitate systems.

The key conceptual bridge is: ODS creep resistance is not governed only by Orowan stress Gb/L ; it also depends on climb, detachment, threshold stress, particle-interface

relaxation, volume fraction, particle morphology, and stability. The Kassner chapter frames this through threshold stress, climb, and detachment models, while Nie's Mg paper shows why shape/orientation can strongly alter effective obstacle spacing and Orowan strengthening efficiency.  kassner-ODS  1-s2.0-S1359646202004979-main

1. Threshold stress vs. Orowan stress

X-axis: Orowan stress,

$$\sigma_{or} = M \frac{Gb}{L}$$

Units: MPa

Y-axis: Experimental threshold stress,

$$\sigma_{th}$$

Units: MPa

Importance:

This is probably the most important baseline plot. It directly tests whether the measured creep threshold scales with classical Orowan strengthening. Many ODS systems show σ_{th} as only a fraction of σ_{or} , often implying climb or detachment control rather than pure bowing.  kassner-ODS

2. Normalized threshold stress vs. normalized particle spacing

X-axis: Normalized interparticle spacing,

$$L/b$$

Dimensionless

Y-axis: Normalized threshold stress,

$$\sigma_{th}/G$$

Dimensionless

Importance:

This collapses alloys with different elastic moduli and Burgers vectors. It allows comparison across Fe-, Ni-, Ti-, Al-, Pt-, and refractory-based ODS alloys.

3. Threshold efficiency map

X-axis: Particle volume fraction,

$$f$$

Units: vol.% or dimensionless

Y-axis: Threshold efficiency,

$$\eta_{th} = \frac{\sigma_{th}}{\sigma_{or}}$$

Dimensionless

Importance:

This identifies whether adding more particles truly improves creep resistance efficiently. A high f but low η_{th} may indicate poor interfacial pinning, coarsening, ineffective particle morphology, or weak particle/matrix interaction.

4. Relaxation factor map

X-axis: Relaxation factor,

$$k_R$$

Dimensionless

Y-axis: Detachment stress ratio,

$$\sigma_d / \sigma_{or}$$

Dimensionless

Importance:

This is the central "interface physics" plot. In the detachment model,

$$\tau_d = \sqrt{1 - k_R^2} \frac{Gb}{L}$$

so lower k_R means stronger particle-dislocation attraction and higher detachment stress. This could be used to classify ODS alloys by interface strength rather than only particle spacing.  kassner-ODS

5. Apparent activation energy vs. threshold efficiency

X-axis: Threshold efficiency,

$$\sigma_{th}/\sigma_{or}$$

Dimensionless

Y-axis: Apparent creep activation energy,

$$Q_{app}$$

Units: kJ/mol

Importance:

High Q_{app} combined with high σ_{th}/σ_{or} suggests thermally activated detachment or interface-controlled creep rather than ordinary lattice diffusion creep. The Kassner chapter notes that activation energies in the threshold regime can exceed lattice self-diffusion.

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6. Creep rate vs. reduced stress master plot

X-axis: Reduced modulus-compensated stress,

$$\frac{\sigma - \sigma_{th}}{G}$$

Dimensionless

Y-axis: Diffusion-compensated creep rate,

$$\dot{\epsilon} \frac{kT}{DGb}$$

Dimensionless

Importance:

This is the classic way to determine whether ODS alloys collapse onto a matrix-controlled creep law after subtracting threshold stress. If they do not collapse, it means additional mechanisms such as interfacial detachment, particle coarsening, grain-boundary sliding, or shape effects are active.  kassner-ODS

7. Particle size vs. creep strength

X-axis: Mean particle radius,

$$r$$

Units: nm

Y-axis: Creep strength at fixed condition, e.g.

$$\sigma(\dot{\epsilon} = 10^{-7} \text{ s}^{-1}, T = 0.7T_m)$$

Units: MPa

Importance:

Tests the predicted optimum particle size. Very small particles may allow easier thermally activated detachment, while very large particles increase spacing and reduce Orowan/detachment stress.  kassner-ODS

8. Interparticle spacing vs. creep strength

X-axis: Mean interparticle spacing,

$$L$$

Units: nm

Y-axis: Creep strength at fixed strain rate and homologous temperature

Units: MPa

Importance:

A direct design plot. Smaller L should generally increase creep resistance, but outliers reveal important effects of particle chemistry, shape, coherency, or interfacial pinning.

9. Volume fraction vs. interparticle spacing

X-axis: Particle volume fraction,

$$f$$

Units: vol.%

Y-axis: Interparticle spacing,

$$L$$

Units: nm

Importance:

This is a processing–microstructure map. It reveals whether a processing route creates many fine particles or fewer coarse particles. It also separates mechanically alloyed ODS systems from precipitation-strengthened or internally oxidized systems.

10. Shape factor vs. creep strength

X-axis: Particle aspect ratio,

$$AR = \frac{\text{major axis}}{\text{minor axis}}$$

Dimensionless

Y-axis: Creep strength at fixed $\dot{\epsilon}$ and T/T_m

Units: MPa

Importance:

Essential for incorporating faceted particles, plate-like dispersoids, and gamma titanium alloys. Nie's paper shows that shape and orientation can dominate strengthening because they change the effective planar obstacle spacing. [1-s2.0-S1359646202004979-main](#)

11. Effective planar obstacle spacing vs. physical particle spacing

X-axis: 3D center-to-center interparticle spacing,

$$L_{3D}$$

Units: nm

Y-axis: Effective slip-plane obstacle spacing,

$$\lambda_{eff}$$

Units: nm

Importance:

This plot captures why rods, plates, faceted particles, and spherical oxides are not equivalent. For prismatic plates, λ_{eff} can become dramatically smaller than expected from simple 3D particle spacing. Nie's analysis showed prismatic plates are much more effective than basal plates, rods, or spheres for blocking basal slip in Mg.

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12. Shape-corrected Orowan stress vs. measured creep strength

X-axis: Shape-corrected Orowan stress,

$$\sigma_{or}^{shape} = M \frac{Gb}{\lambda_{eff}}$$

Units: MPa

Y-axis: Creep strength or threshold stress

Units: MPa

Importance:

This is better than plotting against Gb/L alone. It incorporates particle shape, orientation, and slip-plane intersection geometry. This could become one of the signature plots of Denise's paper.

13. Homologous temperature vs. threshold stress retention

X-axis: Homologous temperature,

$$T/T_m$$

Dimensionless

Y-axis: Normalized threshold stress,

$$\sigma_{th}(T)/\sigma_{th}(RT)$$

Dimensionless

Importance:

This reveals thermal stability of strengthening. Good ODS alloys should retain a large fraction of threshold stress at high T/T_m . Weak systems will collapse because of coarsening, recovery, interface changes, or loss of particle effectiveness.

14. Coarsening resistance map

X-axis: Exposure time,

$$t$$

Units: h

Y-axis: Particle coarsening ratio,

$$r(t)/r_0$$

Dimensionless

Alternative Y-axis:

$$L(t)/L_0$$

Importance:

ODS alloys are valued because oxide particles are thermally stable. This plot directly distinguishes stable nanoscale dispersions from precipitate-strengthened systems that coarsen rapidly.

15. Creep ductility vs. creep strength

X-axis: Creep strength at fixed condition

Units: MPa

Y-axis: Rupture strain or creep elongation

Units: %

Importance:

This is a performance tradeoff map. The best ODS alloys are not just strong; they must retain usable ductility. This plot will help separate brittle high-strength systems from balanced high-temperature structural alloys.

Recommended “signature” plots for the paper

The strongest set would be:

1. σ_{th} vs. σ_{or}
2. σ_{th}/σ_{or} vs. f
3. σ_{th}/G vs. L/b
4. Q_{app} vs. σ_{th}/σ_{or}
5. $(\sigma - \sigma_{th})/G$ vs. diffusion-compensated creep rate
6. r vs. creep strength
7. L vs. creep strength
8. AR vs. creep strength
9. λ_{eff} vs. L_{3D}
10. Shape-corrected σ_{or}^{shape} vs. measured σ_{th}

The most novel contribution would be to introduce a **particle effectiveness index**:

$$\Phi_{ODS} = \frac{\sigma_{th}}{MGb/\lambda_{eff}}$$

where:

- $\Phi_{ODS} \approx 1$: highly effective particle-interface strengthening
- $\Phi_{ODS} \ll 1$: particles exist but are not effectively resisting creep
- Φ_{ODS} can be compared across spherical oxides, faceted particles, rods, and plates

This could be the central unifying metric for the paper.

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